

*Technology Assessment: (Basic def $\rightarrow$ Lenski $\rightarrow$ Horton Hunt $\rightarrow$ Conclusion)
O MR SD C

*** Technology is the organization of knowledge, people and things to accomplish specific practical goals. Sociologist Gerhard Lenski has defined technology as "information about how to use the material resources of the environment to satisfy human needs and desires".
Again, according to Horton and Hunt, "Technology is the use of scientific discoveries to solve practical problems." In conclusion, technology is ~~comprised~~ comprised of the products and processes created by engineers to meet our needs and demands.

*"Many people ~~are~~ people regard technology as simply applied science"
- Do you agree? (People are wrong $\rightarrow$ why they are wrong)

Many people ^{indeed} regard technology as simply applied science. In their view, scientist produce knowledge and then technologists turn it into important products and devices, such as, computers and spacecraft. However, we see science did not begin to be systematically incorporated into our production system until the middle of the nineteenth century. If technology is simply "applied science" we would have to conclude that there was no technology before that.

In fact, quite complex and sophisticated technology was needed to build Pyramids of Egypt, The Great Wall of China and the ancient irrigation systems of India and Sri Lanka. Such knowledge was based on

craft (rather than science,) with knowledge being slowly accumulated, applied, and passed on from one generation to the next. So, it can be concluded that technology is not just applied science.

* The importance of Science and Technology: (Dev. Countries → Effect of science → Impact →)

Most people would agree that science and technology are of great importance in the world today. Most highly developed countries spend 2 to 3 percent of their GDP on science technology. These large sums tell that decision-makers in government and industry are strongly convinced of the importance of developing science and technology.

Science can alter the entire conception of us and our place in the universe. ~~This ranges from~~ The most famous instance of this was the series of events known as the scientific Revolution. During this turbulent times, ~~all the~~ some of the most important inventions ~~like~~ and discoveries ^{de the solar system to} like the origin of the whole universe were made. Whatever the future holds, it seems certain that science will play a major part in shaping our view of the universe and ~~as~~ of ourselves.

In the 19th century, science-based technology began to ~~form~~ transform whole industries. In this century, it has made warfare far more dangerous thanks to the nuclear weapons. We have also seen the rapid onslaught of computerization and telecommunications ^{technology} as well as modern pharmaceuticals which can cure diseases that used to ^{be} ~~be~~ pests. These have changed virtually all aspects of our lives.

Impact of Politics on Science and Technology [and war]

Politics has played a crucial part in Science, Technology and Society movement. The earliest efforts arose out of the revolution in Russia (1917) and establishment of ^{the} ~~a~~ socialist state, ^{Soviet Union}. One part of Marx and Lenin's theory was the materialist interpretation of history. In other words, they argued that all significant changes were ~~caug~~ caused by the productive forces. of course, this places technology at the very heart of historical change.

Later on, this Marxist view ~~was~~ of science ~~were~~ in ~~told~~ became known to the scholars in the West through a conference on history of science, called "Science at the Crossroads." There, the ^{through his papers} ~~so~~ Soviet ~~to~~ delegation and historian Boris Hessen argued ^{that} Newton was led to ~~advances in the technologies~~ that address certain sorts of problems because their solution would lead to advances in technologies that were important to the dominant social forces of the time; navigation, ^{mining} and development of weaponry.

Although Soviet Union collapsed in 1992, it managed to open the eyes of many a thinkers as the state financed large scientific and technological projects and did not leave developments to ~~chang~~ chance.

* Impact of War

War also had a major impact on the analysis of the role of science and technology in society of particular importance were:

i) The development of atomic bomb in 1942 and ^{its use made} later many scientists of the project regret their involvement.

ii) The next 50 years after the initial development of the atomic bomb, the rest of the world saw the production of huge numbers of ^{US and SU} nuclear weapons in the Cold War arms race between US and SU and lived with the prospect of total destruction.

iii) ~~Governments realized that for their countries to progress, they had to~~

ii) A distinguished Scientist, Vannevar Bush, was asked to report on a suitable plan after World War II. He recommended the ~~setting up~~ ^{what is now called the} establishment of the now National Science Foundation. He also wrote ~~the~~ ^a report which advocated the setting up of a national policy concerned with science.]

iv) ~~Governments realized that for their countries to progress, they had to support scientific research and technical technological development. As a result of this, governments began planning for science and technology. After the war, spending on science was growing exponentially, but this could not continue, as eventually~~

science would consume all of the government's budget. Governments had to make choices about what to fund and develop a criteria to help them make choices. As such, two key questions ~~are~~ arose; ~~which~~ whether governments or scientists are best ~~to~~ equipped to decide what should be funded in science, and how "success" is to ~~be~~ judged.

* Technology Assessment :

~~In short~~, Technology assessment refers to checking the boons and banes of any technology as well as ^{how to increase} ~~de~~ ^{increasing} the ^{positive effects} ~~positivity~~ while reducing the negative effects of said technology.

Science, ~~and~~ technology and society has come to embrace ~~another an~~ ^{the} area of study, ~~know that~~ technology assessment and concerned with the impact of large-scale technologies on society. As technology becomes bigger and more complicated it follows that the impacts are felt far and wide if something goes wrong, even if it is a minor problem. Researchers in this field look at emerging technologies and try to ensure that positive outcomes are exploited and negative outcomes are avoided. In short, it goes over:

- i) Economic feasibility
- ii) Social acceptability
- iii) Engineering constraints
- iv) Previous experience

Hunting and Gathering → Beginning of Human Life → Nomadic, reliance on readily available food and fibres, mostly kinship ties, jobs were assigned based off of gender and family background

Horticulture → About 12k years ago → More settled, development of agriculture and limited technology, less nomadic and more focus on household objects, sticks and tools.

Agrarian → About 5k years ago → Larger, more stable settlements, improved technology, increased crop yields, minor yet significant technological developments, reliance on physical power of cattle and humans, social structure is more defined, free time to think because of surplus production, creation of artifacts and art objects.

Industrial → 1750 - 1860 → Reliance on mechanical power and new source of energy - centralized workplaces, economic independence, formal education, specialized workforce

Post industrial → 1960s → Reliance on services, especially the processing and control of information, expansion of middle class, more psychological jobs such as teaching, advertisement firms, public relations etc, declining organized work force, increase in interest groups, conflict could persist

Postmodern → 1970s → High technology, mass consumption of consumer goods and media images, cross cultural integration, technologically sophisticated, various theoretical perspectives,

*Technology Culture:

~~Technological~~ Technology culture refers to an attitude of individuals in a given cultural environment. The spirit of inquiry, the degree of acceptance of the right to question and be questioned is considered fundamental to the development of technological temper. It calls upon one to seek the "how", "what" and "why" of everything that goes on in the society. The existence of a technology culture is complementary to the initiative taken by a country in the introduction of productive forces which can lead to technological development.

* Roadblocks for development of technology: (+AEF)

The relationship between culture and technology ~~are~~ have been identified as one of the reasons of some of the negative elements in the socio-economic systems prevailing in the regions which are believed to have acted as road blocks for development of technology. Below are some of those road blocks or negative elements in the society which probably are more significant than others among the different causes of uneven progress in developing a technology culture:

a. Traditional Value System and Orthodoxy:

[Distrust of new technologies is deeply rooted in most cultures with old value systems because people in general don't want to change]. The fears of changes that usually follow introduction of new technology results from the belief that these changes are likely to be adverse. New technology, be it the product of local development, a transfer from abroad, or some combination the two may not agree with existing traditional values. Scientific and technological changes can undercut systems of belief. [It will seem that there is always a fear in a society steeped in orthodoxy that something important (e.g. religious belief, family values, social equilibrium etc) would be lost as a result of the new technology.]

b. Ancient Habits of Resignation:

Many people in developing regions continue to believe that there is a limited quantity of resources ^{and attempt to expand # resources} through some form of technological innovation likely to result in a smaller share for them. Increased productivity can increase the share for everyone, yet in many countries people do not believe this and are resigned to the old modes of production.

c. Stratified and Exclusive Societies:

Many countries in the developing regions still retain stratified social structures that resist change and continue with rigid class or caste systems. Meritocracy is not encouraged and the excluded section of the society, ~~the~~ wrongly called the "lower class", is resigned to the rule of the ruling class and thus the urge to excel ~~at~~ is lost and spirit of competitiveness is discouraged.

d. Highly Centralized Bureaucratic Decision-Making Systems Discouraging

Diffusion of Ideas and Technologies:

Almost all cultures in the world have been borrowers of technology throughout history. Useful technology spread, either through migration of populations or by diffusion of techniques to neighbouring

population both within and outside geo-political. Unfortunately, however, in the developing regions there have been periods when some ruling classes or powerful centralized ~~bureaucracies~~ ^{bureaucracies} intentionally decided to resist science and technology interchange and diffusion process thus stopping local innovation activities. ~~Wrong~~ ^{Wrong} decisions taken by such centralized bodies negatively influenced vast populations.

e. Education System Discourages Nurturing of Questioning Minds and has Inadequate Focus on Tertiary Education:

[Since the latter half the last century the products of scientific research and technological innovation have given rise to a "knowledge society"] Plans for education reforms often lead to proposals to import computers, involve foreign experts/consultants, find health care diagnostic laboratories etc. However, [creating an environment that inspires local talent to innovate requires additional efforts in terms of openness to cross-border exchanges of teachers and students, creating atmosphere for interdisciplinary dialogue and debate and the deliberate promotion of questioning minds and creativity in the school curricula. Such determined and purposeful initiatives are seen only in a few countries of the developing regions. [In most cases, the students during their primary and secondary stage go through a poorly designed assessment procedure and find it convenient to depend on rote learning. 12 years of rote learning can easily kill creativity in any young mind]

F. Poor Connectivity with World Systems and Inadequate Communication

Tools:

[The ~~use of~~ use of communication tools ~~have~~ has been critical in enabling some societies to develop complex organizations, which in turn create the correct social environment for technology to grow. These tools are a medium for the spread of technology and ideas, which help develop an innovative culture]. Societies which did not welcome early adoption of communication tools failed to adopt technological advances and were unable to make them available to the population. Even before the advent of modern hi-tech devices, earlier communication tools had lowered the cost and increased the speed and numbers of exchanges that encourages innovation. [However, only those who have "plugged in" can take advantage of this worldwide interconnectedness. This also requires broad literacy, investment in information and communication technologies and shift towards openness to the free exchange of ideas.]

Technology Determinism and its Criticism

Technology determinism is the belief that technology is the agent of social change which is the notion that technological development is autonomous with respect to society. In other words, it implies that technology shapes society but it is not reciprocally influenced. Technology moulds society and change our behaviours. It exists outside the society, but at the same time influences social change. In technology determinism, technology is seen as the most significant determinant of the nature of society. According to Marx and Smith, ~~one hand~~, hard technological determinists impute agency to technology, and imply that technological development is inescapable and inevitable. Marx said, "The windmill gives you a society with feudal lord; the steam mill with industrial capitalist." ~~On the other~~

However, according to some sociologists, technology determinism is unsatisfactory. Some criticisms include:

i) Technologies do not follow some predetermined course. Research and development are the significant determinant of the sorts of technologies to develop. Its development is also dependent on a broad range of social, political and economic factors. It is clear that culture play a role in shaping the history of technological development, and that societies do not develop along a unique, fixed path.

ii) Soft determinists recognize human agency, and that changes in history are due to various and complex social, economic and cultural factors.

ii) Some say that technology does not determine human action but rather human actions shape technology. The way a technology is used cannot be understood without understanding how that technology is embedded in its social context. SCOT, or the social construction ^{of technology} approach, is a helpful approach for understanding the innovation of particular artifacts.

iv) Actor-network theory proposes a network linking human beings and non-human entities. A technological system ties inextricably together both material artifacts and human beings and ties together with technology.

* Ethics: (Ethics → Root Word → Actions)

Ethics is a set of beliefs about right and wrong behaviour within a society. "Ethics" is derived from the Greek term "ethos" which refers to the notions of custom, habit, behaviour and character. In other words, ethics is the specific and descriptive set of actions that are more concerned with the results when not done.

* Difference between Ethics and Morality: (Ethics, Morality → OACTPE)

Ethics is a set of beliefs about right or wrong behaviour within a society. Meanwhile, morality morals are the welfare principles enunciated by the wise people, based on their experience and wisdom, and morality is concerned with principles and practices of morals. Although ethics and morality are often used interchangeably in everyday discourse we draw ~~some~~ ^{the following} important distinctions between the two terms:

Topic	Morality	Ethics
i) Origin of word (O)	Latin word	"Mores" — Greek word "ethos"
ii) Application spectrum (A)	More general and prescriptive — based on customs and traditions	Specific and descriptive. It is a critical reflection on moral
iii) Concerned (C)	More concerned with the results of wrong action when done	More concerned with the right action when not done
iv) Thrust (T)	Thrust is on judgement and punishment in the name of God or by laws	Thrust is on influence, education, training through codes, guidelines and connection

Topic

Morality

Ethics

- v) Priority (P) — In case of conflict between the two, morality is given top priority because damage is more. It is more common and basic.
- vi) Example (E) — Character flaw, corruption, extortion and crime.
- Less serious, hence second priority only. But relevant today, because of complex interactions in the modern society.
- Notions or beliefs about manners, tastes and customs towards laws.

* Ethical Engineering

* Engineering Ethics:

Engineering ethics is -

- i) the study of moral issues and decisions confronting individuals and organizations engaged in engineering
- ii) the study of related questions about the moral ideals, character, policies and relationships of people and corporations involved in technological activities.

* Ethical Theories:

(Ethical ideas were continually refined during the course of history. Many great thinkers have turned their attention to ethics and morals. The thinking of these philosophers is especially important

For our study of engineering ethics as ^{these} ~~that~~ moral principles are universal, regardless of their origin, and are applicable even in secular settings. The ethical theories can be classified into the following categories:

- i) Utilitarianism: (Utilitarianism holds those actions as good that serve to maximize human well being)
- ii) Duty Ethics: ~~The main~~ (This type of ethics believes that moral duties are fundamental)
- iii) Right Ethics: (Right ethics holds that people have fundamental rights that others people have a duty to respect.)
- iv) Virtue Ethics: (Virtue ethics is interested in determining what kind of people we should be.)

#Utilitarianism / Criticism of Utilitarianism (i) / (ii) / (iii)
Tries to balance / How will you solve? : (Utilitarianism → Dam → WIPP → WIPP analysis → Criticism)

iii) (Utilitarianism holds those actions as good that serve to maximize human well-being. The emphasis in utilitarianism is not on maximizing the well-being of individual rather on maximizing the well-being of the society as a whole. It is called collective approach.) For example: building of dam. Dams often lead to great benefit to society by providing stable supplies of drinking water, flood control, and recreational opportunities. However, these benefits often come at the expense of people who live in areas that will be flooded by the dam as they are required to find new ho shelters. Utilitarianism tries to balance the needs of society with

the needs of the individual, with an emphasis on what will provide the most benefit to the most people. However, it can be seen that in the example that sometimes what is best for everyone may be bad for a particular individual or group of individuals. Another example of this problem is:

Waste Isolation Pilot Plant:

~~a near~~ ~~Carls~~ - near carlsbad, New Mexico

- designed to be a permanent repository for nuclear waste
- a system of tunnels bored into underground salt beds which are considered to be very stable by geologists
- people oppose this facility as transportation occurs across highways and accidents can cause problems for people living near these routes

An analysis of WIPP using utilitarianism might indicate that the disposal of nuclear wastes is a major problem hindering the implementation of many useful technologies, including medicinal uses of radioisotopes and nuclear generation of electricity. Solution of this waste disposal problem will benefit society by providing slight improved health care and more plentiful electricity. The slight potential for adverse health effects for individuals living near the transportation routes is far outweighed by the overall benefits to the society. So, WIPP should be allowed to open.] iii

As this example demonstrates, the utilitarian approach can be seen to ignore the needs of individuals, especially if these needs seem relatively insignificant. Besides, it is impossible to know exactly what the consequences of an action are and when human agencies are involved as subject of experiments it is impossible to determine

all potential outcomes. So, maximizing the benefit to society involves guesswork and the risk that the best guess might be ~~was~~ wrong.

Despite these objections, utilitarianism is a valuable tool for ethical problem solving, providing one way of looking at engineering ethics case.) ii

Cost-Benefit Analysis: (Cost-Benefit Analysis $\rightarrow$ Pitfalls/Criticism)

One tool often used in engineering analysis, especially when trying to determine whether a project makes sense, is cost-benefit analysis. Fundamentally, ~~this type of analysis is just an application~~ In cost-benefit analysis. Only those projects with the highest ratio of benefits to costs will be implemented. This principle is similar to the utilitarian goal of maximizing overall good.

The pitfalls of cost-benefit analysis involves:

- i) While it is often easy to predict the costs for most projects, the benefits that are derived from them are harder to predict and to assign a dollar value.
- ii) The ratio can't take into account many of the more subjective aspects of a decision like scenic wilderness or loss of an endangered species.
- iii) It is sometimes impossible to determine if those who stand to reap the benefits are also those who paid the cost.

i) *Virtue Ethics : (Virtue ethics $\rightarrow$ Virtue $\rightarrow$ Ideology $\rightarrow$ Questions)

Virtue ethics is interested in determining what kind of people we should be. Virtue is often determined as moral and goodness. In virtue ethics, actions are considered right if they support good character traits. Virtue ethics focuses on words such as responsibility, honesty, competence and loyalty. Other relevant virtues might include trustworthiness, fairness, caring, citizenship and respect.

The main ideology of virtue ethics is that personal morality cannot be separated from professional morality. If a behaviour is virtuous in ^{any} individual's personal life, the behaviour is virtuous in his/her professional life as well. And as such, we can use virtue ethics in our engineering careers by answering the following questions:

- i) Is this action honest?
- ii) Will this action demonstrate loyalty to my community, employers and/or client?
- iii) Have I acted in a responsible fashion?

The answers to these questions makes the proper course of action obvious and helps to solve a lot of problems.

*Code of Ethics: (Engineering why code is imp → 6 codes)

Engineering is an important and learned profession. As members of this profession, engineers are expected to exhibit the highest standards of honesty and integrity. Engineering has a direct and vital impact on the quality of life for all people. Accordingly, the services provided by engineers require honesty, impartiality, fairness, and equity and must be dedicated to the protection of the public health, safety and welfare. Engineers must perform under a standard of professional behavior that requires adherence to the highest principles of ethical conduct.

Engineers, in the fulfillment of their professional duties, shall:

- i) Hold paramount the safety, health, and welfare of the public.
- ii) Perform services only in the areas of their competence.
- iii) Issue public statements only in an objective and truthful manner.
- iv) Act for each employer or client as faithful agents or trustees.
- v) Avoid deceptive acts.
- vi) Conduct themselves honorably, responsibly, ethically, and so lawfully so as to enhance the honor, reputation, and usefulness of the profession.

* Engineers Code 1: (~~If life is endangered~~ life $\rightarrow$

The first code of ethics for engineers states that, "Engineers shall hold paramount the safety, health and welfare of the public."

The rules of practices of this code includes:

- a. Engineer IF engineers' judgement is overruled under circumstances that endangers life or property, they shall notify their employer or ~~client~~ client and such other authority as may be appropriate.
 - b. Engineers shall approve only those engineering documents that are in conformity with applicable standards.
(If endangered, contact authority)
 - c. Engineers shall not reveal facts, data, or information without prior consent of the client or employer except as authorized or required by this law or code.
(maintain quality)
 - d. Engineers shall not permit the use of their name or associate in business ventures with any person or firm that they believe is engaged in fraudulent or dishonest enterprise.
(not reveal facts)
 - e. Engineers shall not aid or abet the unlawful practice of engineering by a person or firm.
(not associate with frauds)
 - f. Engineers having knowledge of any alleged violation of this Code shall report thereon to appropriate professional bodies and, when relevant, also to public authorities, and cooperate with the proper authorities in furnishing such information or assistance as may be required.
(no unlawful acts)
- (contact authority when code is violated)

*Engineers Code 2:

The second code of ethics for engineers states that, "Engineers shall perform services only in the areas of their competence." The rules of practices of this code includes:

- a. Engineers shall undertake assignments only when qualified by education or experience in the specific technical fields involved.
- b. Engineers ^(only when qualified) shall not affix their signatures to any plans or documents dealing with subject matters in which they lack competence, nor to any plan or document not prepared under their direction and control.
- c. Engineers ^(not signing where they did not participate) may accept assignments and assume responsibility for coordination of an entire project and sign and seal the engineering documents for the entire project, provided that each technical segment is signed and sealed only by the qualified engineers who prepared the segment.

*Code 5 ^{(let people s}
^(giving credit where credit is due)

* Code 5:

The fifth code of ethics for engineers states that, "Engineers shall avoid deceptive acts." The rules of practices of this code include

- a. Engineers shall not falsify their qualifications or permit or misrepresentation ~~of~~ misrepresentation of their or their associates' qualifications. They shall not misrepresent or exaggerate their responsibility in or for the subject matter of prior assignments. Brochures or other presentations incident to the solicitation of employment shall not misrepresent pertinent facts concerning employers, employees, associates, joint ventures, or past ~~acc~~ accomplishments.
- b. ^(Not falsify their actions) Engineers shall not offer, give, solicit, or receive, either directly or indirectly, any contribution to influence the award of a contract by public authority, or which may be reasonably construed by the public as having the effect or intent of influencing the awarding of a contract. They shall not offer any gift or other valuable consideration in order to secure work. They shall not pay a commission, percentage, or brokerage fee in order to secure work, except to a bona fide employee or bona fide established commercial or marketing agencies retained by them.

^(Not falsify others' actions)

*Code 3:

The third code of ethics for engineers states that, "Engineers shall issue public statements only in an objective and truthful manner."

The rules of practices of this code includes:

- a. Engineers shall be objective and truthful in professional reports, statements, or testimony. They shall include all ~~a~~ relevant and pertinent information in such reports, statements, or testimony, which should bear the date indicating when it was current.
(~~say what~~ (document and do it truthfully))
- b. Engineers may express publicly technical opinions that are founded upon knowledge of the facts and competence in the subject matter.
(say if what went wrong)
- c. Engineers shall issue no statements, criticisms, or arguments on technical matters that are inspired or paid for by interested parties, unless they have prefaced their comments by explicitly identifying the interested parties on whose behalf they are speaking, and by revealing the existence of any interest the engineers may have in the matter.
(~~do~~ don't be bribed)

*Code 4:

The fourth code of ethics for engineers states that, "Engineers shall act for each employer or client as faithful agents or trustees."

The rules or practices of this code includes:

- a. Engineers shall disclose all known or potential conflicts of interest that could influence or appear to influence their judgement on the quality of their services.
- b. Engineers shall ^(inform about problems) not accept compensations, financial, or otherwise, from more than one party for services on the same project, or for services pertaining to the same project unless the circumstances are fully disclosed and agreed to by all interested parties.
- c. Engineers shall not ^(not work for two parties at once) solicit or accept financial or other valuable consideration, directly or indirectly, from outside agents in connection with the work for which they are responsible.
- d. Engineers in public service as members, advisors, or employees of a governmental or quasi governmental body or department shall not participate in decisions with respect to services solicited or provided by them or their organizations in private or public engineering practice. ^(not be biased)
- e. Engineers shall not solicit or accept a contract from a governmental body on which a principal or officer of their organization serves as a member. ^(not to take jobs from same employees in different setting)

*Impact of IT on ^{Standard of Living} Productivity^a (GDP → Rate of Change → Downturn)

The standard of living varies greatly among groups within a country as well as from nation to nation. The most widely used measurement of the material standard of living is gross domestic product (GDP) per capita. National GDP represents the total annual output of nation's economy. Overall, industrialized nations tend to have a higher standard of living than developing countries.

In the United States, as in most developed countries, the standard rate of living has been improving over time. However, its rate of change varies as a result of business cycles that affect prices, wages, employment levels, and the production of goods and services. Major disasters — such as earthquakes, hurricanes, tsunamis and wars — can negatively impact the standard of living.

The worst economic downturn in U.S. history occurred during the Great Depression, when the GDP declined by about 50 percent from 1929 to 1932 and by 1932, the unemployment rate ^{of 2007} had reached 25 percent. By way of comparison, during the latest recession, in the U.S. the growth rate declined by 6.8 percent during the fourth quarter of 2008 and the U.S. unemployment rate hit a peak of 10.2 percent in 2009. Needless to say, these were all thanks to the rapid development of IT sectors.

* Measuring productivity after IT investment: (Productivity $\rightarrow$ Factory + Service)

Productivity is defined as the amount of output produced per unit of input and it is measured in many different ways. They are:

i) Productivity in Factory:

Productivity in a factory might be measured by the numbers of labor hours it takes to produce one item.

ii) Productivity in Service Sector:

Productivity in a service sector company might be measured by annual revenue an employee generates divided by their annual salary.

* Factors that have increased the prevalence of Telework: (Defⁿ $\rightarrow$ ABTGEF)

Telework, also known as telecommuting, is a work arrangement in which an employee works away from the office. The factors that increased the prevalence of Telework are as follows:

- i) Advances in technology that enable people to communicate and access the internet from anywhere. **A**
- ii) The increasing numbers of broadband connections in homes and retails. **B**
- iii) High levels of traffic congestion. **T**
- iv) Rising gasoline prices. **G**
- v) Growing concern over the effects of automobile CO₂ emissions. **E**
- vi) Frequent demands of the hi scarce and highly trained workers about more flexible work arrangements, flex hours and the ability to occasionally work from home. **F**

* Advantages and Disadvantages from the perspective of an employee

In telework, an employee uses various forms of electronic communication, including texting, email, audio and video calling as and online chat. As such, the advantages and the disadvantages they receive include:

Advantages: DCTWA

- i) People with disabilities who otherwise find public transportation and office accommodations as a barrier to work may now be able to join the workforce. (Office Accommodation) D
- ii) Teleworkers avoid long and stressful commutes and gain time for additional work or personal activities. (Additional Work) C
- iii) Telework can reduce the need for employees to take time off to stay home to care for a sick family member. (Monetary) (Misc) T
- iv) Teleworkers have an opportunity to experience an improved work-family balance. (Monetary/Family) W
- v) Telework reduces ad hoc work requests and disruptions from fellow workers. (Increased Productivity) A

Disadvantages: P I OR HC

- i) Some employees are unable to be productive workers away from the office
- ii) Teleworkers may suffer from isolation and may not really feel like they are "part of the team" I
- iii) Workers who are out of sight tend to be out of mind and as such their contributions may not be fully recognized and credited. R
- iv) Teleworkers must guard against working too many hours per day because work is always there. H
- v) The cost of the necessary equipment and communication services can be considerable if the organization does not cover these. C

* Advantages and Disadvantages from the perspective of an Employer

In telework, an employee . . . and online chat. As such, the advantages and disadvantages the employers face include:

Advantages: OMA&C

- i) As more employees telework, there is less need for office and parking space; this can lead to lower costs. (Office Accommodation) O
- ii) Allowing employees to telework can improve morale and reduce turnover. (Morale / Family) M
- iii) Telework allows for continuity of business operations in the event of a local or national disaster and supports national pandemic preparedness planning. (Additional Work) A
- iv) There may be an actual gain in worker productivity. (Increased Productivity) G
- v) Telework can decrease an organization's carbon footprint. (Misc) C

Disadvantages: SDMPCL

- i) Allowing teleworkers to access organizational data and systems from remote sites create potential security issues. S
- ii) Informal, spontaneous meetings become more difficult if not impossible. D
- iii) Managers may have a harder time monitoring the quality and quantity of the work performed by teleworkers, wondering, for instance, if they really "put in a full day." M
- iv) Increased planning is required by managers to accommodate and include teleworkers. P
- v) There are additional costs associated with providing equipment, services and support for people who work away from the office. C
- vi) Telework increases the potential for lost or stolen equipment. L

What is Digital Divide and why does it exist/persist everywhere?

(Defⁿ → Difference)

"Digital Divide" is a term used to describe the gap between those who do and don't have access to modern information and communication technology; such as smart phones, personal computers, ~~at~~ internet etc. ~~The digital~~ Digital Divide exists both in between and within countries. The reasons it persists/exists everywhere are:

i) Difference in Age Groups:

Digital Divide exists because among age groups. In the United States, Hispanics and African Americans ^{fewer, seniors} ~~seniors~~ citizens are least likely to have ~~less~~ internet connections compared to the youth.

ii) Separation in Economic classes:

In the United States, Hispanics and African Americans adults living in poor households are least likely to have internet access.

iii) Contrast in Living Location:

There ~~are~~ is a significant divide between people who live in cities and those ~~who~~ in rural areas. There are roughly 2.4 billion internet users worldwide, but the distribution of those users vary greatly from region to region. In Bangladesh, only 38.1% of households have internet access, ^{where} of which 29.7% of said percentage is rural and 63.4% is of urban areas.

As long as such issues exist ^{worldwide} globally, so will the global plague of digital divide.

*Bridging Digital Divide: (Defⁿ → Mobile Phone → Other Advantage)

"Digital Divide" is a term used to describe the gulf between those who do and don't have access to modern information and communication technology; cell phones, smart phones, personal computers, internet etc. Digital divide exists both between and within countries. However, one of the biggest tools to bridge this divide has been mobile or cell phones.

Some industry observers identify the increasing use of cell phones as ~~first~~ an important first step in bridging the digital divide in many countries. The rapid and widespread use of cell phones have resulted in an increased investment in the infrastructure required to support wireless communications. In addition, as cellphone use has spread, financial institutions and other organizations have built applications capable of accepting text-based input to process user transactions and store cash or credits on users' phones.

Other advantages include: CPELAT

- i) Cell phones come in wide range of capabilities and costs, but are cheaper than personal computers. C
- ii) Cellphones are more portable and convenient than the smallest of laptops. P
- iii) Cell phones come with an extended battery life. E
- iv) There is almost no learning curve required to use cell phone. L
- v) Basic cell phones require no costly or burdensome applications. A
- vi) There are essentially no technical-support challenges to overcome when using a cell phone. T

*Impact of IT in healthcare costs: (Everything is expensive $\rightarrow$ Telehealth + Mobile applications)

The rapidly rising cost of health care is one of the 21st century's major challenges. The development and use of new medical technology account for much of the increase in healthcare spending per person in excess the time of general inflation excessively.

To really gain control over soaring healthcare costs, patients awareness must be raised and technology costs must be managed more carefully. In the meantime, however, the improved use of IT in the healthcare industry can lead to significant cost reductions in a number of ways. Some are:

- i) Tele health employs electronic information processing and telecommunications to support at-a-distance health care, provide professional and patient health-related training and support healthcare administration. This technology reduces the need for patients to travel for treatment and allows healthcare professionals to serve more patients in a broader geographic area.
- ii) Thousands of mobile applications are available to improve patient access to healthcare information and to enable doctors to keep a close watch on patients' condition. Appointment and prescription reminders, medication and vital sign tracking and diet and weight monitoring are common applications based on the sending of text messages to the patient, the healthcare professional or even a monitoring computer.

Thus, the influence of IT in the healthcare has truly made diagnosis and treatment a lot more efficient and a lot less expensive.

*Telemedicine + Types: (Telemedicine → ^{Different types} 3 sentences)

Telemedicine is the component of tele health that provides medical care to people at a location different from the healthcare providers. This technology reduces the need for patients to travel for treatment and allows healthcare professionals to serve more patients in a broader geographic area. There are three main basic forms of telemedicine:

i) Store-and-forward: (Store data → Forward to doctor)

This type of telemedicine involves acquiring data, sound, images and video from a patient and then transmitting everything to a medical specialist for later evaluation. This type of monitoring does not require the presence of the patient and care providers at the same time. Yet, having access to such information can enable healthcare professionals to recognize problems and intervene with remote patients before high-risk situations become life threatening.

ii) Live Telemedicine: (Communication with specialist)

Live telemedicine requires the presence of patients and healthcare providers at different sites at the same time and often involves a videoconference link between the two sites. Oil companies and the like are increasingly relying on live telemedicine to connect to a nurse or EMT on an oil platform to emergency physicians at a major medical centers.

iii) Remote Monitoring: (Tracking vital signs)

Remote monitoring, also called home monitoring involves the regular transmission of the ~~and~~ on going measurement of an individual's vital signs (temperature, blood pressure, heart rate etc) and other health measures ~~and~~ to a health care provider. Patients who have chronic diseases often don't recognize early ^{warning} signs that indicate an impending health crisis. While the patient may not be aware of the potential dangers, a physician could be alerted to this potentially life-threatening development before a health crisis occurs. It is estimated that 2.8 million people worldwide used some form of home monitoring device 2012.

*Causes of Globalization:

Globalization means integration of countries through commerce, transfer of technology, and exchange of information and culture. Anthony Giddens describes globalization as "the intensifications of worldwide social relations which link distinct localities in such a way that local happenings are shaped by events occurring miles away." The causes of Globalization include:

i) Improved Communications:

The development of communications technologies such as internet, email and mobile phones have been vital to the growth of globalization because they help MNCs to operate throughout the world. The development of satellite TV channels such as Sky and CNN have also provided worldwide marketing avenues for the concept and products of globalization.

ii) Improved Transport:

The development of refrigerated and container transport, bulk shipping and improved air transport has allowed the easy mass movement of goods throughout the world. This assists globalization.

Free Trade Agreement:

MNCs and rich capitalist countries have always promoted global free trade as a way of increasing their own wealth and influence. International organizations such as the World Trade Organization and the IMF also promote free trade.

iv) Global Banking:

Modern communication technologies allow vast amounts of capital to flow freely and instantly throughout the world.

v) The Growth of the MNCs:

The rapid growth of big MNCs (such as Microsoft, McDonalds and Nike) is both a cause and a consequence of globalization, while the investment of MNCs in farms, mines and factories across the world is a major part of globalization, globalization also allows MNCs to produce goods and services and to sell products on a massive scale throughout the world.

* Conceptualize Globalization and Human Rights:

Globalization:

Globalization means integration of countries through commerce, transfer of technology and exchange of information and culture. Anthony Giddens describes globalization as "the intensifications of worldwide social relations which link distinct localities in such a way that local happenings are shaped by events occurring miles away. It includes acting together and interacting economies through trade, investment, loan, development schemes and capital across countries. These flows include knowledge, science, technology, skills, culture, information and entertainment, besides direct human resource, telework and outsourcing.

Human Rights:

Human rights are defined as all those rights which are essential for the protection and maintenance of dignity of individuals and create conditions in which every human being can develop his personality to the fullest extent ~~may be termed as human rights~~. Human beings are national beings and they by virtue of their being human possess certain basic and inalienable rights which ~~by~~ are commonly known as human rights. Human rights become operative with the birth of an individual. Human rights being ^{the} birth rights are inherent in all the individuals irrespective of their caste, religion, sex and nationality. Human rights are also sometimes referred to as fundamental rights, basic rights, inherent rights, natural rights and birth rights. Today, the concept of human rights include civil or political rights or public liberties, economic, social and cultural needs and it is the state's responsibility to protect human rights.

Summarize
*Bhopal / and its accidents:

In 1983, there were 14 plants in India manufacturing chemicals, pesticides, and other hazardous products. The Bhopal plant had a license to make Methyl isocyanate-based pesticides. In November 1984, they had decided to close down the plant as for quite some years before the production rate was going down. The cumulative effects ^{of the factors} that caused the tragedy in Bhopal on December 3, 1984 are summarized in the following:

- i. The safety manual of Union Carbide prescribed that the MIC tanks were to be filled up to 60%. But the tanks were reported to have been filled up to 75%.
- ii. The safety policy prescribed that an empty tank should be available as a stand-by in case of emergency. But the emergency tank was also filled to its full capacity.

- iii. The storage tanks should be refrigerated to make the chemical less reactive. But here the refrigeration was shut down as an economy measure. This raised the temperature of gas stored.
- iv. The plant was shut down for maintenance two months earlier. The worker who cleaned the pipes and filters connected to the tanks and closed the valves, was not trained properly. He did not insert the safety disks to prevent any possible leakage of the gas. This led to the build up of the temperature and pressure in the storage tanks.
- v. When the gas started leaking out, the operators tried to use the vent gas-scrubber that was designed to reduce the extinguishing gas but that scrubber was also shut down.
- vi. There was a flame tower that was designed to burn off the gas escaping from the scrubber. That was not also in working condition.
- vii. The workers finally tried to spray water up to 100 feet to quench the gas (which was water soluble). But the gas was escaping from the chimney of 120 - feet high.
- viii. The workers were not trained on safety drills or emergency drills or any evacuation plan.

*Determinants of Human Rights:

Human rights are defined as all those rights which are essential for the protection and maintenance of dignity of individuals and create conditions in which every human being can develop his personality to the fullest extent. According to the model of Poe and Tate, a government's decision to respect human rights is determined by the following ~~six~~ six areas:

i) The level of Democracy:

In Poe and Tate's model, human rights repression is a tool for governments to solve conflicts. If a government is threatened, it reacts by repressing human rights; imprisonment, torture, killings etc. In democracies, repressive governments can be ousted from office in elections. Moreover, the characteristic freedom in democracies allow citizens to publicise and denounce abusive actions.

ii) The level of economic development and its growth:

Regarding economic conditions, Poe and Tate argue that human rights abuses should decrease with economic development, yet increase with economic growth. The better the economic situation, measured by the quantity and quality of goods and service available, the less scarce are the goods and consequently the potential for conflict and thus repression decreases. Since economic growth is the engine of economic development, it should improve human rights practices in a country. However, growth usually involves winners and losers, and temporarily, increases the inequality of wealth and income and consequently social and political tensions rises and probability of abusive actions increases.

iii) Population size and its growth rate:

Population growth increases the scarcity of the products and resources. Furthermore, rapid population growth increases the percentage of younger people in terms of the total population, which as a group is generally more destabilizing. Therefore, both population size and population growth may increase the probability of repressive ~~government~~ governmental actions.

iv) The prevalence of a military and/or leftist regime:

Military regimes depend on the power of the armed forces, and conflicts are frequently solved by military orders and hierarchy. Here, military regimes are most likely to control a country through abusive actions and disrespect of human rights whenever possible. Similarly, "leftist regimes" which are defined as socialist or communist government that "do not allow effective electoral competition with non-socialist opposition", which is based on the Marxist-Leninist theory that there is a need for the dictatorship of the proletariat, are most likely to show respect for human rights.

v) British cultural influence:

The colonial experience has prejudiced the shape of present postcolonial political culture, and that the British colonial experience was associated with post colonial development of democracy. British cultural influence in this sense, therefore, assumed to involve greater respect for human rights.

vi) Recent experience with international and/or civil war:

Poe and Tate point out that experience with external or internal war increases the probability of human rights violations. As Benkhart puts it, "civil liberties disappear in even the most democratic of countries come wartime."

*Impact of Globalization on Human Rights:

Globalization means integration of countries through commerce, transfer of technology and exchange of information and culture while human rights are defined as all those rights which are essential for the protection and maintenance of dignity of individuals and create conditions in which every human being can develop his personality to the fullest extent.

i) Widening the gap between the rich and the poor:

A primary effect of economic globalization is the ~~exa~~ exacerbation of gaps between the rich and the poor. Statistics show that this gap is widening both within and between nations and that, in many countries, poverty is both increasing by numbers and by depth. The weakness of international rules, bad policies and weak governance in developing countries and corporate practices priorities short-term profit over long-term human development are undermining the capacity of

poor countries and poor people to benefit from international trade. In many cases, economic liberalization has been accompanied by greater inequality and people are left trapped in utter poverty. The Human Development Report of 1997 revealed that poor countries and poor people too often find their interests neglected because of globalization.

ii) Violation of fundamental human rights:

Economic globalization has resulted in the violation of the internationally proclaimed human rights underlined by the Universal Declaration of Human Rights (UDHR). The rights most likely to be violated by the TNCs include non-discrimination, women rights, life, liberty and physical integrity of the person, civic freedoms, employee rights, child labour, slavery, forced and bonded labour, right to food, health, education and housing and lastly, the environmental rights. On the other hand, in their drive for profits, the TNCs have been restructuring their operation on a global scale.

iii. Threatening the states sovereignty and democracy:

Constraints on state sovereignty have come about through the process of globalization, which largely is a form of Westernism. Some people also have termed this process as "Americanization" or "McDonaldization" due to tremendous participation of TNCs from the US. This process naturally affects non-Western societies

more than Western ones. Globalization, therefore, has created a situation where the role and importance of nation-state is becoming irrelevant. In addition, the globalization of recent decades was never a democratic choice by the peoples of the world. The process has been business driven, by business strategies and tactics, for business ends. Thus, globalization has also steadily weakened democracy and state's sovereignty.

*Concept of Appropriate Technology:

Identification, transfer, and implementation of most suitable technology for a set of new situations is called appropriate technology. The term "appropriate" is value based and it should ensure fulfillment of human needs and protection of the environment. Depending on the availability of resources, physical conditions, capital opportunities, costs and the human value system which includes their traditions, beliefs and religions etc the appropriateness is to be determined. In other words, technology that are assessed and are convenient for usage are known as appropriate technology.

*Technology Transfer:

Technology Transfer ~~is the~~ refers to the process of moving technology to a new setting and implementing it there. It may also mean moving the technology applications from laboratory to the field/factory or from one country to another. Technology includes hardware (machines and installations) and the techniques (technical, organizational and managerial skills and procedures). This transfer is affected by governments, organizations, universities and MNCs.

*Advantages and Disadvantages of MNCs :

Organizations who have established business in more than one country are called multinational corporation. The headquarters are in home and the business is extended in many host countries. The advantages and disadvantages of such organizations include: